

HEYUAN, China

WMO: 592930

Lat: 23.7935N

Lon: 114.7309E

Elev: 61

StdP: 100.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	5.6	-7.5	2.0	10.8	-4.3	2.6	10.7	8.9	8.8	8.0	9.2	3.0	40	0.304

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	35.2	25.9	34.4	25.8	33.6	25.7	27.7	31.6	27.2	31.3	26.8	30.9	2.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.6	29.3	26.2	21.8	28.9	25.8	21.2	28.7	88.8	31.8	86.8	31.5	85.0	31.2	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.4	4.5	2.7	37.1	1.5	0.8	1.6	37.7	0.7	38.2	-0.2	38.7	-1.3	39.3		
			0.0	28.2	1.7	1.4	-1.2	29.2	-2.2	30.0	-3.2	30.8	-4.5	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	13.5	15.3	18.2	22.6	25.9	27.9	29.2	29.0	27.8	24.7	20.0	15.0
	DBStd	6.43	4.19	4.95	4.14	3.35	2.49	2.02	1.63	1.72	2.05	2.53	3.54	3.61
	HDD10.0	32	14	11	1	0	0	0	0	0	0	0	1	4
	HDD18.3	466	157	107	55	8	0	0	0	0	0	1	23	115
	CDD10.0	4583	123	160	256	378	492	537	596	590	534	456	302	159
	CDD18.3	1974	8	23	51	135	233	287	337	332	284	198	74	12
	CDH23.3	19033	27	97	236	822	2016	3031	3987	3840	2989	1553	392	44
CDH26.7	7181	1	7	29	186	656	1138	1754	1663	1188	482	76	2	
Wind	WSAvg	1.9	2.1	2.0	1.9	1.7	1.7	1.7	1.8	1.7	1.9	2.0	2.1	2.1
Precipitation	PrecAvg	1927	61	73	141	221	281	384	228	250	154	43	42	49
	PrecMax	2747	415	230	556	547	682	992	483	521	422	184	182	181
	PrecMin	1274	0	1	44	36	77	145	50	68	53	0	0	0
	PrecStd	410	77	60	106	122	139	198	131	143	81	46	48	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	27.7	29.1	31.9	34.3	35.1	36.5	36.3	35.1	33.6	30.7	26.7
		MCWB	19.0	19.8	21.6	23.1	25.4	26.1	26.0	26.1	25.6	24.4	22.4	20.5
	2%	DB	23.8	26.1	27.4	30.1	32.6	34.1	35.2	35.1	34.0	31.7	28.8	24.6
		MCWB	17.6	19.1	20.5	22.8	24.5	25.9	26.0	26.0	25.3	23.4	21.3	18.4
	5%	DB	21.9	24.4	26.0	28.8	31.4	33.0	34.3	34.2	33.1	30.6	27.2	22.7
		MCWB	16.2	18.5	19.8	22.4	24.3	25.7	26.0	25.9	25.1	22.5	20.1	17.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	20.0	22.7	24.4	27.5	30.2	31.8	33.2	33.1	32.1	29.4	25.7	20.8
		MCWB	15.5	18.0	19.5	22.2	24.0	25.7	26.0	25.8	24.6	21.8	19.1	15.4
	0.4%	WB	20.1	21.7	23.1	25.4	27.5	28.1	28.2	28.2	27.5	25.8	23.7	22.2
		MCDB	23.6	25.5	26.4	28.5	31.1	31.5	32.1	32.4	31.5	30.2	28.2	24.8
	2%	WB	18.8	20.7	22.2	24.5	26.3	27.3	27.6	27.6	26.8	25.1	22.9	20.1
		MCDB	22.2	23.9	25.3	27.5	29.9	31.1	31.7	31.8	31.0	29.1	26.7	23.2
Mean Daily Temperature Range	5% DB	WB	17.5	19.7	21.4	23.8	25.7	26.8	27.1	27.1	26.3	24.5	22.0	18.4
		MCDB	20.9	22.9	24.5	27.0	29.2	30.6	31.5	31.3	30.3	28.3	25.6	21.3
		WB	16.2	18.6	20.5	23.1	25.2	26.4	26.7	26.7	25.8	23.8	20.8	16.5
	5% WB	MCDB	19.6	22.0	23.6	26.1	28.5	30.0	30.9	30.7	29.6	27.5	24.2	20.0
Clear-Sky Solar Irradiance	taub	0.586	0.652	0.786	0.774	0.626	0.593	0.568	0.629	0.657	0.676	0.574	0.554	
	taud	1.741	1.657	1.451	1.514	1.829	1.912	1.974	1.805	1.726	1.640	1.829	1.834	
	Ebn at Noon	663	653	593	616	713	731	749	705	672	626	670	672	
	Edh at Noon	207	240	309	295	214	195	184	217	231	241	188	183	
All-Sky Solar Radiation	RadAvg	2.88	2.89	3.03	3.23	3.85	4.03	5.01	4.77	4.47	4.09	3.35	3.06	
	RadStd	0.47	0.50	0.44	0.58	0.64	0.58	0.46	0.43	0.32	0.50	0.43	0.48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page